REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Curved Mirrors Practice Questions

Grade 10 Advanced | Module 16 Lesson 2 | Curved Mirrors

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: f = R/2; 1/f = 1/do + 1/di; m = hi/ho = −di/do. Concave f positive; convex f negative; real di positive.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 9–18.

P01. A concave mirror can form a real image when the object is:

A. At every possible position B. Always between mirror and focus
C. Beyond its focal point D. Behind the mirror only

P02. A convex mirror forms images of real objects that are:

A. Real inverted and enlarged B. Real upright and diminished
C. Virtual inverted and enlarged D. Virtual upright and diminished

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P03. For a spherical concave mirror, focal length is approximately:

A. Half the radius of curvature B. Twice the radius of curvature
C. Equal to object height D. Independent of curvature

P04. A negative magnification means the image is:

A. Always virtual B. Inverted
C. Always larger D. At infinity

P05. A real object at the centre of curvature of a concave mirror has an image that is:

A. Virtual upright and enlarged B. Real upright and diminished
C. Real inverted and equal in size D. Virtual inverted and equal in size

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P06. A spherical concave mirror has radius of curvature 24 cm. Calculate its focal length.

A. 12 cm B. 24 cm
C. 6 cm D. -12 cm

P07. A concave mirror has focal length +10 cm. An object is 30 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

A. 30 cm B. 15 cm
C. 7.5 cm D. -15 cm

P08. A convex mirror has f = -10 cm and object distance do = +20 cm. Find the signed image distance.

A. -13.33 cm B. -3.333 cm
C. -6.667 cm D. 6.667 cm

P09. A mirror forms a real image 12 cm in front when the object is 24 cm in front. Calculate its signed magnification.

A. -1 B. -0.25
C. 0.5 D. -0.5

P10. A mirror has magnification −0.50 for an object 6 cm tall. Find the signed image height.

A. -3 cm B. -6 cm
C. -1.5 cm D. 3 cm

P11. Define principal axis of a spherical mirror.

P12. Define focal point of a concave mirror.

P13. Define focal length.

P14. Why are convex mirrors useful for vehicle observation?

P15. What does positive signed image distance mean under the stated mirror convention?

P16. A spherical concave mirror has radius of curvature 36 cm. Calculate its focal length.

P17. A spherical concave mirror has radius of curvature 48 cm. Calculate its focal length.

P18. A concave mirror has focal length +15 cm. An object is 45 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

P19. A concave mirror has focal length +20 cm. An object is 60 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

P20. A convex mirror has f = -15 cm and object distance do = +30 cm. Find the signed image distance.

P21. A convex mirror has f = -20 cm and object distance do = +40 cm. Find the signed image distance.

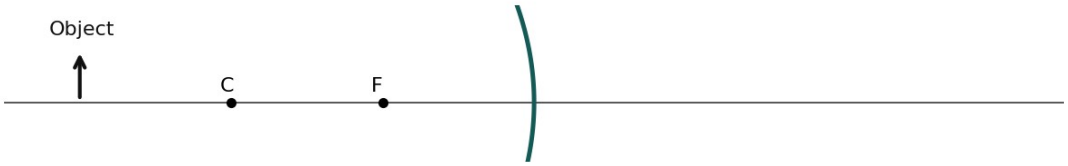
P22. A mirror forms a real image 18 cm in front when the object is 36 cm in front. Calculate its signed magnification.

P23. A mirror forms a real image 24 cm in front when the object is 48 cm in front. Calculate its signed magnification.

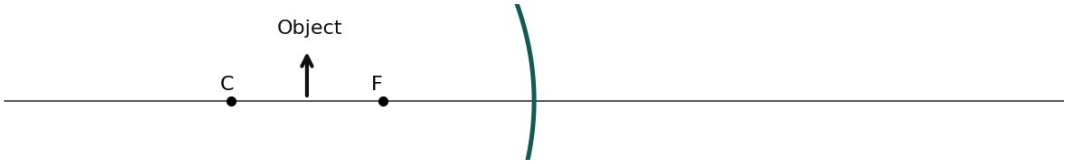
P24. A mirror has magnification −0.50 for an object 9 cm tall. Find the signed image height.

P25. A mirror has magnification −0.50 for an object 12 cm tall. Find the signed image height.

P26. Draw a ray diagram for an object beyond C of a concave mirror. State the image properties.



P27. Draw a ray diagram for an object between F and C of a concave mirror.



P28. A concave mirror is used as a magnifying shaving mirror. Where should the face be relative to F? Explain.

P29. A student claims that any image larger than the object must be virtual. Evaluate the claim.

P30. Why should a ray diagram be checked against the sign of calculated magnification?